

HDOC Day-38 — String Operations

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Algorithm

1. Start.
2. Declare a character array to store the string.
3. Read the string using `fgets()`.
4. Repeatedly display the menu:
 - Count spaces, digits, and special characters.
 - Count frequency of a given character.
 - Toggle case of each character.
 - Exit.
5. Read the user's choice.
6. Use a `switch` statement:
 - **Choice 1:** Call a function that examines every character and counts spaces, digits, and special characters.
 - **Choice 2:** Read the character to search for and call a function to count its occurrences.
 - **Choice 3:** Call a function that converts lowercase letters to uppercase and uppercase letters to lowercase.
 - **Choice 4:** Exit the program.
 - Otherwise, display "Invalid choice!".
7. Continue until choice 4 is selected.
8. Stop.

Test-Case Table

Test Case	Operation	Input	Expected Output
1	Count spaces/digits/special	Hello 123!	Spaces = 1, Digits = 3, Special = 1
2	Count frequency	String: banana, Character: a	Frequency = 3
3	Toggle case	Hello World	hELLO wORLD
4	Count spaces/digits/special	C Programming @ 2026	Spaces = 3, Digits = 4, Special = 1
5	Count frequency	String: programming, Character: m	Frequency = 2
6	Toggle case	AbC123@xYz	aBc123@XyZ
7	Invalid menu choice	Choice = 6	Invalid choice!
8	Exit	Choice = 4	Program terminates

C Program

```
#include <stdio.h>

void countCharacters(char str[]);
void characterFrequency(char str[]);
void toggleCase(char str[]);

int main()
{
    char str[200];
    int choice;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Count spaces, digits and special\ncharacters\n");
        printf("2. Count frequency of a given character\n");
        printf("3. Toggle the case of each character\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                countCharacters(str);
                break;

            case 2:
                characterFrequency(str);
                break;

            case 3:
                toggleCase(str);
                break;

            case 4:
                printf("Exiting program...\n");
                break;
        }
    }
}
```

```

        default:
            printf("Invalid choice! Please enter 1 to
4.\n");
        }

    } while (choice != 4);

    return 0;
}

/* Function to count spaces, digits and special characters */
void countCharacters(char str[])
{
    int i = 0;
    int spaces = 0, digits = 0, special = 0;

    while (str[i] != '\0')
    {
        if (str[i] == ' ')
        {
            spaces++;
        }
        else if (str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if (!((str[i] >= 'A' && str[i] <= 'Z') ||
            (str[i] >= 'a' && str[i] <= 'z')) &&
            str[i] != '\n')
        {
            special++;
        }

        i++;
    }

    printf("\nNumber of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

/* Function to find frequency of a character */
void characterFrequency(char str[])
{
    char ch;

```

```

int i = 0, count = 0;

printf("Enter the character to search: ");
scanf(" %c", &ch);

while (str[i] != '\0')
{
    if (str[i] == ch)
    {
        count++;
    }

    i++;
}

printf("Frequency of '%c' = %d\n", ch, count);
}

/* Function to toggle case */
void toggleCase(char str[])
{
    int i = 0;

    printf("String after toggling case: ");

    while (str[i] != '\0')
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            printf("%c", str[i] - 32);
        }
        else if (str[i] >= 'A' && str[i] <= 'Z')
        {
            printf("%c", str[i] + 32);
        }
        else
        {
            printf("%c", str[i]);
        }

        i++;
    }
}

```

Execution

```
[aaryanrajput@aaryans-macbook ~ % nano day38.c
[aaryanrajput@aaryans-macbook ~ % clang day38.c -o day38.c
[aaryanrajput@aaryans-macbook ~ % ./day38.c
Enter a string: hello guys

----- MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 1

Number of spaces = 1
Number of digits = 0
Number of special characters = 0

----- MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 2
Enter the character to search: 3
Frequency of '3' = 0

----- MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 4
Exiting program...
aaryanrajput@aaryans-macbook ~ %
```